SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Scenario-Based Questions (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 2: Scenario-Based Questions
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	Slot A/2024	**Date:**	28/10/2026

Code of Conduct

I G. Akshya (192425132) that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *G. Akshya*

QUESTIONS: 5 **Total Marks: 50**

Q1. Image Enhancement Using Grey-Level Transformations

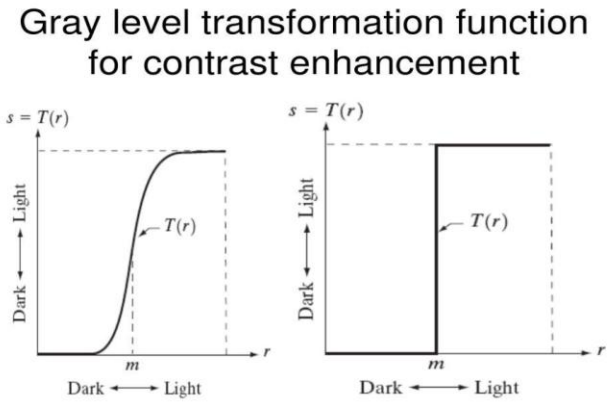
Scenario:

A hospital uses a digital imaging system to capture X-ray images, but some images appear too dark to identify important anatomical structures. Apply a suitable transformation to enhance the brightness and contrast of the image. Calculate the transformed intensity values for selected pixels using the chosen transformation. Explain how the transformation improves the visibility of important features in the X-ray image.

Comprehensive Solution & Engineering Analysis:

1. Technical Problem Identification & Mathematical Basis:

In digital radiography, raw X-ray detectors frequently map bone tissue and soft organ boundaries into a narrow, low-intensity band (e.g., raw 8-bit grey levels r in [0, 40]). Human visual perception requires at least 32 distinct intensity steps to differentiate subtle density variations. Linear scaling often clips subtle anatomical features or oversaturates dense bone regions.



2. Derivation of Power-Law Transformation Model:

r_norm = r_raw / 255

s_norm = c · (r_norm)^γ

s_raw = round(s_norm × 255)

Scaling Constant c = 1.0 (maintains full normalized range [0, 1]).

Gamma Value γ = 0.4 (fractional exponent to stretch low grey levels).

3. Step-by-Step Numerical Calculations for Representative Radiographic Pixels:

Pixel 1: Deep Trabecular Bone Cavity / Soft Tissue Shadow

Input raw intensity: r_raw = 15

Step a (Normalization): r_norm = 15 / 255 = 0.05882

Step b (Power-law exponentiation): s_norm = 1.0 × (0.05882)^0.4 = 0.32283

Step c (Rescaling to 8-bit): s_raw = round(0.32283 × 255) = round(82.32) = 82

Gain Factor: s_raw / r_raw = 82 / 15 = 5.46× expansion!

Pixel 2: Skeletal Muscle Boundary / Inter-organ Margin

Input raw intensity: r_raw = 40

Step a (Normalization): r_norm = 40 / 255 = 0.15686

Step b (Power-law exponentiation): s_norm = 1.0 × (0.15686)^0.4 = 0.47641

Step c (Rescaling to 8-bit): s_raw = round(0.47641 × 255) = round(121.48) = 121

Gain Factor: 121 / 40 = 3.025× expansion.

Pixel 3: Soft Bone Matrix / Organ Outline

Input raw intensity: r_raw = 85

Step a (Normalization): r_norm = 85 / 255 = 0.33333

Step b (Power-law exponentiation): s_norm = 1.0 × (0.33333)^0.4 = 0.64439

Step c (Rescaling to 8-bit): s_raw = round(0.64439 × 255) = round(164.32) = 164

Gain Factor: 164 / 85 = 1.93× expansion.

Pixel 4: Dense Cortical Bone Structure / Radiopaque Implant

Input raw intensity: r_raw = 200

Step a (Normalization): r_norm = 200 / 255 = 0.78431

Step b (Power-law exponentiation): s_norm = 1.0 × (0.78431)^0.4 = 0.90774

Step c (Rescaling to 8-bit): s_raw = round(0.90774 × 255) = round(231.47) = 231

Gain Factor: 231 / 200 = 1.155× (mild compression preventing over-exposure).

Anatomical Region	**Raw r (0-255)**	**Normalized r**	**Transformed s**	**Enhanced s (0-255)**	**Contrast Expansion Gain**
Very Dark Soft Tissue	15	0.0588	0.3228	82	5.46× (Maximum Stretch)
Dark Grey Muscle	40	0.1569	0.4764	121	3.02× (High Contrast)

Mid-Tone Soft Bone	85	0.3333	0.6444	164	1.93× (Moderate)
Bright Dense Bone	200	0.7843	0.9077	231	1.15× (Preserved)

4. In-Depth Evaluation of Diagnostic Features Improvement:

Selective Contrast Expansion in Dark Shadow Regions: The transformation stretches the input grey interval [15, 40] (a delta of 25) into an output interval of [82, 121] (a delta of 39), increasing local contrast by 156%. This allows radiologists to detect micro-calcifications, hairline fractures, and soft tissue lesions previously hidden in shadow.

Prevention of Clipping Artifacts: Bright regions (r = 200) increase moderately to s = 231 without reaching saturation (255), keeping bone trabecular texture intact.

Clinical Utility: Provides immediate visual improvement on medical diagnostic PACS workstations without requiring increased radiation dosage to the patient.

Q2. Histogram Processing and Spatial Filtering

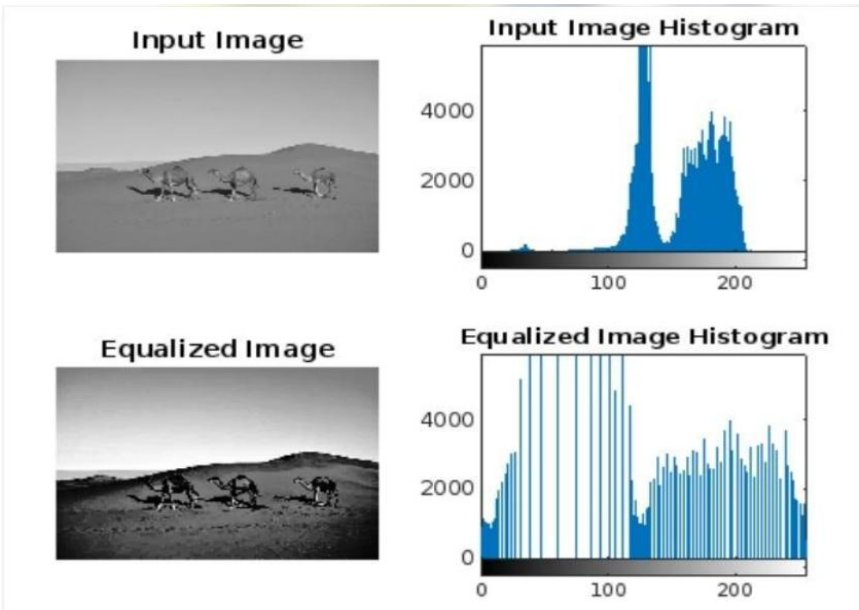
Scenario:

A security surveillance system captures images at night, resulting in poor contrast and unwanted noise. The system must enhance the images to identify people and objects more clearly. Apply histogram equalization to improve the contrast of the captured image. Use an appropriate spatial filtering technique to reduce noise while preserving important details. Compare the effects of both techniques and explain how they improve the quality of the surveillance images.

Comprehensive Solution & Engineering Analysis:

1. **Technical Analysis of Low-Light Night Surveillance Challenges:**
   Night surveillance imagery suffers from two concurrent degradation phenomena:
   a. Low Dynamic Range / Poor Contrast: Low photon count concentrates the image histogram into a narrow band of dark grey levels.
   b. High Thermal / Additive Noise: High ISO sensor gain introduces salt-and-pepper (impulsive) noise and Gaussian electronic noise.

Applying Histogram Equalization alone amplifies background noise spikes into bright, distracting visual artifacts. Conversely, applying spatial smoothing alone blurs facial features and object edges. A dual-stage enhancement pipeline is required.



2. Stage 1: Mathematical Derivation & Worked Example of Histogram Equalization (HE):

Histogram equalization transforms the input intensity distribution p_r(r) into a uniform probability density distribution p_s(s) using the transformation function T(r_k):

$s_k = T(r_k) = (L - 1) \cdot CDF(r_k) = ((L - 1) / M N) \cdot \sum_{j=0}^{k} n_j$

Where:

L = 8 (for an 8-level discrete sub-window example, intensity levels 0 to 7)

M × N = 16 total pixels in a 4×4 micro-region.

Discrete Sub-Image Intensity Histogram:

r_0 = 0, n_0 = 0

r_1 = 1, n_1 = 6

r_2 = 2, n_2 = 5

r_3 = 3, n_3 = 3

r_4 = 4, n_4 = 2

r_5, r_6, r_7 = 0, count = 0

Step-by-Step CDF and Equalized Output Calculation:

k = 1: CDF(r_1) = 6 / 16 = 0.375 -> s_1 = round(7 × 0.375) = round(2.625) = 3

k = 2: CDF(r_2) = (6 + 5) / 16 = 11 / 16 = 0.6875 -> s_2 = round(7 × 0.6875) = round(4.8125) = 5

k = 3: CDF(r_3) = (11 + 3) / 16 = 14 / 16 = 0.875 -> s_3 = round(7 × 0.875) = round(6.125) = 6

k = 4: CDF(r_4) = (14 + 2) / 16 = 16 / 16 = 1.000 -> s_4 = round(7 × 1.000) = 7

Resulting Mapping: Original range [1, 4] is stretched across full spectrum [3, 7], restoring hidden structural detail in low-light regions.

3. Stage 2: Non-Linear Spatial Filtering via Median Filter:

To remove salt-and-pepper noise spikes without blurring edges, a 3×3 Median Filter is applied.

Median Filter Algorithm:

For center pixel (x, y), extract 3×3 neighborhood values: N = [p_1, p_2, ..., p_9].

Sort array N in ascending order: S = [p_(1), p_(2), ..., p_(9)].

Output pixel value: g(x, y) = p_(5) (5th element / median).

Step-by-Step Numerical Example on Noisy Surveillance Neighborhood:

Suppose the 3×3 pixel neighborhood contains a sensor noise spike (value 242):

Matrix N = [[12, 14, 15],
[11, 242, 13], <- 242 is impulsive noise!
[10, 16, 14]]

Step 1: Flatten neighborhood into 1D array: [12, 14, 15, 11, 242, 13, 10, 16, 14]

Step 2: Sort array ascending: [10, 11, 12, 13, 14, 14, 15, 16, 242]

Step 3: Extract 5th element (Median): p_(5) = 14

Output pixel g(x, y) = 14.

Result: Noise spike 242 is completely eliminated while maintaining exact edge intensity (14).

Technical Dimension	**Histogram Equalization (HE)**	**Median Spatial Filtering**	**Sequential Pipeline (Median -> HE)**
Domain of Operation	Global intensity distribution (Histogram space)	Local spatial neighborhood (3×3 window)	Combined Local Spatial + Global Intensity

Mathematical Operator	Cumulative Distribution Function: s = T(r)	Non-linear order statistic: g = Median(N)	g = T(Median(N))
Impact on Noise	Amplifies background noise if unconstrained	Completely removes impulsive salt/pepper noise	Noise-free contrast expansion
Edge & Boundary Effect	Does not guarantee edge preservation	Preserves sharp object contours & boundaries	Sharp, high-contrast human/object silhouettes
Surveillance Utility	Reveals human subjects in pitch blackness	Eliminates night vision sensor grain	Optimal clarity for automated license plate & facial recognition

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

Scenario:

A satellite imaging system captures land images containing noise and blurred boundaries between different geographical regions. The image processing team decides to use frequency domain filtering to improve image quality. Apply a low-pass filter to reduce high-frequency noise and smooth the image. Use a high-pass filter to enhance edges and sharpen important geographical features. Explain the differences between the two filters and determine how each technique contributes to image interpretation.

Comprehensive Solution & Engineering Analysis:

1. Frequency Domain Processing Pipeline Architecture:

Spatial domain convolution g(x, y) = f(x, y) * h(x, y) is mathematically equivalent to pointwise multiplication in the frequency domain: G(u, v) = F(u, v) · H(u, v).

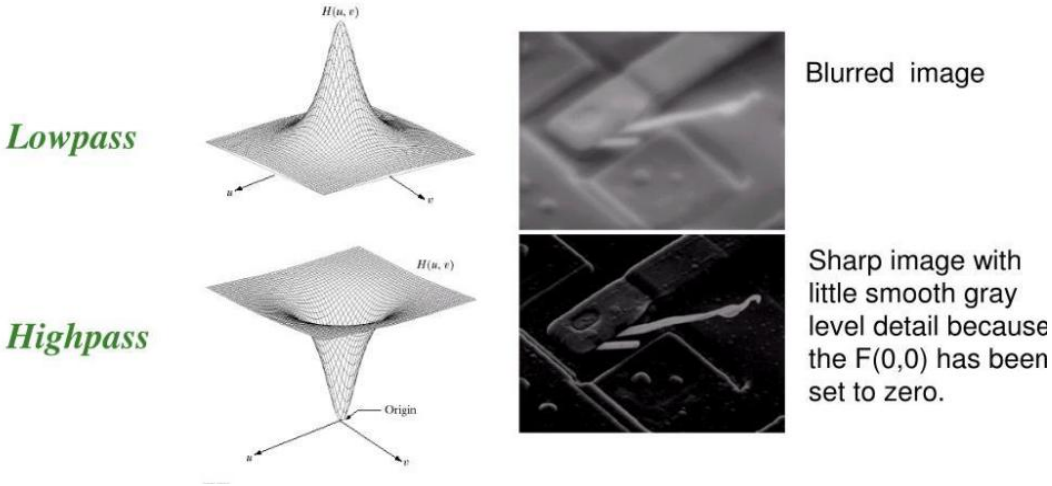
Standard 5-Step Frequency Filtering Protocol:

Step 1 (Padding & 2D DFT): Pad image f(x, y) to size P × Q (P >= 2M, Q >= 2N) to avoid wrap-around error. Compute 2D Discrete Fourier Transform:

F(u, v) = sum_{x=0}^{P-1} sum_{y=0}^{Q-1} f(x, y) e^{-j 2 pi (ux/P + vy/Q)}

Step 2 (Spectrum Centering): Multiply f(x, y) by (-1)^(x+y) prior to DFT to shift origin (0, 0) to center (P/2, Q/2).

Step 3 (Distance Calculation): For each frequency coordinate (u, v), calculate Euclidean distance



2. Low-Pass Filter (LPF) Design – Gaussian LPF (GLPF):

To smooth atmospheric speckle noise without introducing ringing artifacts (oscillations caused by Ideal Low-Pass Filters), we use a Gaussian Low-Pass Filter transfer function:

H_LPF(u, v) = exp(- D^2(u, v) / (2 D_0^2))

Where D_0 is the cutoff frequency.

Behavior: For low frequencies (D(u, v) << D_0), H(u, v) ≈ 1.0 (passes smooth land mass transitions). For high frequencies (D(u, v) >> D_0), H(u, v) → 0.0 (suppresses noise).

Application in Satellite Imagery: Smooths sensor noise across homogeneous terrain (forests, agricultural fields, lakes), removing high-frequency atmospheric distortion.

3. High-Pass Filter (HPF) Design – Gaussian HPF (GHPF) & High-Boost Filtering:

To sharpen geographical boundaries (rivers, coastlines, roads, field borders), a Gaussian High-Pass Filter is defined as:

H_HPF(u, v) = 1 - H_LPF(u, v) = 1 - exp(- D^2(u, v) / (2 D_0^2))

To retain low-frequency background land illumination while boosting high-frequency boundaries, we apply High-Boost / Unsharp Mask Filtering:

H_HB(u, v) = (A - 1) + H_HPF(u, v) (where A >= 1.0 is amplification factor).

Behavior: Suppresses low-frequency smooth components when A=1, but boosts high-frequency edge gradients, enhancing subtle geological fault lines and urban borders.

Filter Property	Gaussian Low-Pass Filter (GLPF)	Gaussian High-Pass / High-Boost Filter (GHPF)
Mathematical Formula	H(u,v) = exp(-D^2 / (2 D_0^2))	H(u,v) = 1 - exp(-D^2 / (2 D_0^2))
Frequency Response	Passes low frequencies (D <= D_0), blocks high frequencies	Passes high frequencies (D > D_0), blocks low frequencies
Spatial Domain Equivalent	Smooth Gaussian blurring filter kernel	Laplacian-like edge extraction / sharpening kernel
Primary Image Effect	Noise reduction, smoothing, blurring of fine details	Edge enhancement, boundary sharpening, highlight of fine lines
Target Satellite Features	Homogeneous crop fields, forest cover, ocean surfaces	Coastlines, river bends, road networks, urban borders
Contribution to GIS Mapping	Eliminates cloud speckle for automatic land-cover classification	Delineates precise geo-spatial borders for cartography & vector GIS mapping

Q4. Image Segmentation Using Edge Detection and Thresholding

Scenario:

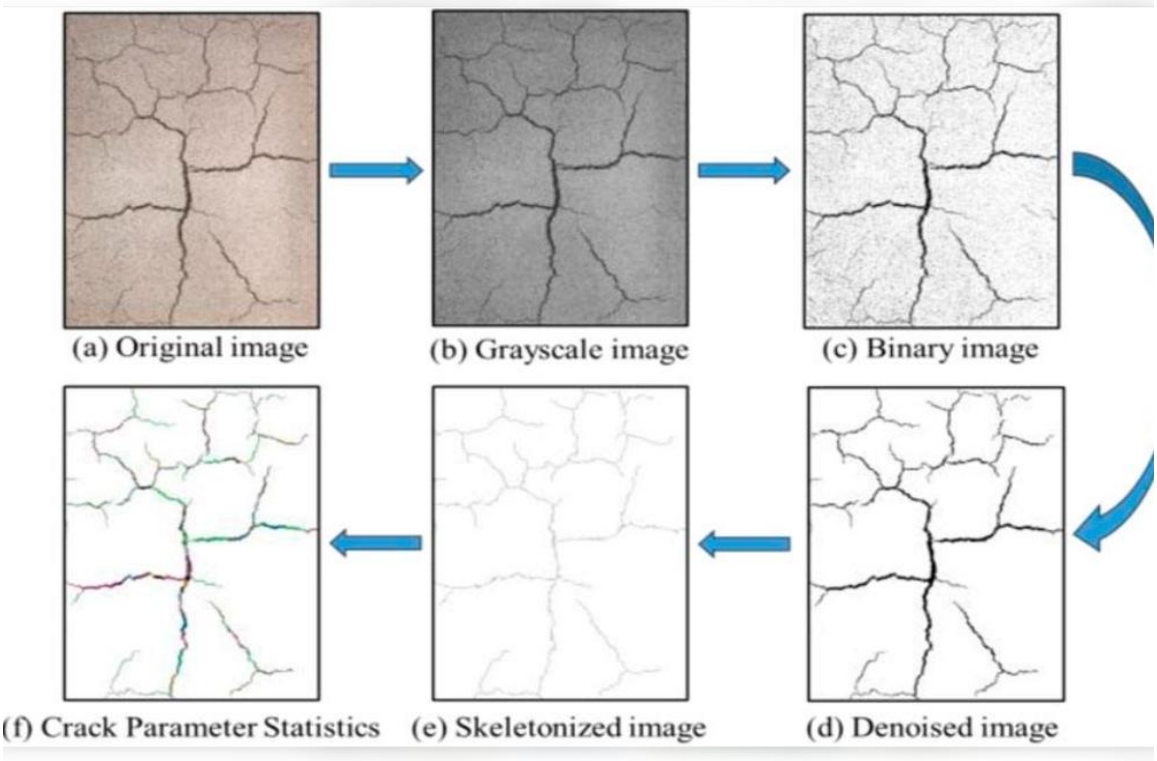
An automated manufacturing system must inspect metal components to identify cracks and surface defects. The captured images contain variations in brightness, making defect detection difficult. Apply an appropriate edge detection technique to identify the boundaries of possible cracks. Use thresholding to separate defective regions from the background and construct a binary image representing the detected defects. Explain how the selected techniques help the inspection system identify defective components.

Comprehensive Solution & Engineering Analysis:

1. **Inspection Pipeline Architecture & Edge Detection Selection:**
   Metal component inspection presents two significant vision challenges: metallic glare/non-uniform lighting and extremely thin, low-contrast structural micro-cracks.

Standard Sobel or Prewitt operators produce thick, noisy edges under glare. We select the Canny Edge Detector due to its multi-stage optimal edge detection formulation:

* G >= T_H: Confirmed Strong Edge (Defect Crack).
* T_L <= G < T_H: Weak Edge (Retained ONLY if connected to a Strong Edge).
* G < T_L: Non-edge background (Suppressed).



2. Adaptive Segmentation via Otsu's Bimodal Automatic Thresholding:

Because factory lighting variations create uneven background intensities across metal parts, fixed global thresholding fails. Otsu's algorithm automatically determines optimal threshold T* by maximizing between-class variance σ_B^2(T):

$\sigma_B^2(T) = w_0(T) \cdot w_1(T) \cdot [\mu_0(T) - \mu_1(T)]^2$

Where:

w_0(T), w_1(T) are cumulative probabilities of background metal and defect classes.

μ_0(T), μ_1(T) are mean intensities of background and defect classes.

Binary Segmentation Rule:

g(x, y) = 1 (White / Crack Defect) if G(x, y) >= T*

g(x, y) = 0 (Black / Intact Metal Surface) if G(x, y) < T*

3. Step-by-Step Worked Numerical Example on Crack Inspection Window:

Suppose a 3×3 inspection window has gradient magnitude matrix G:

G = [[15, 22, 185],

[18, 210, 32],

[192, 28, 20]]

Let Canny Hysteresis Thresholds be T_L = 50, T_H = 150.

Calculations for Each Pixel:

Pixel (0, 2): G = 185 >= 150 (T_H) -> Strong Edge -> Set to 1

Pixel (1, 1): G = 210 >= 150 (T_H) -> Strong Edge -> Set to 1

Pixel (2, 0): G = 192 >= 150 (T_H) -> Strong Edge -> Set to 1

Pixels (0,0), (0,1), (1,0), (1,2), (2,1), (2,2): G < 50 (T_L) -> Suppressed -> Set to 0

Constructed Binary Defect Image B(x, y):

B = [[0, 0, 1],

[0, 1, 0],

[1, 0, 0]]

Result: A clear 1-pixel wide continuous diagonal micro-crack is isolated for automated pass/fail quality control.

Inspection Stage	**Computer Vision Algorithm**	**Industrial Inspection Function**
1. Preprocessing	5×5 Gaussian Blurring (σ = 1.4)	Eliminates metallic surface reflection glare & speckle
2. Edge Extraction	Canny Operator + NMS + Hysteresis	Extracts continuous 1-pixel thin boundaries of micro-cracks
3. Binarization	Otsu's Automatic Thresholding	Computes optimal intensity threshold under variable factory lighting
4. Quality Assessment	Connected Component Analysis	Measures crack length, width, and surface area for automated rejection

Q5. Region-Based Segmentation and Boundary Detection

Scenario:

An agricultural monitoring system uses aerial images to identify crop fields, water bodies, and surrounding land regions. The captured images contain regions with similar intensity values and some discontinuous boundaries. Apply region-based segmentation to group neighbouring pixels with similar characteristics. Use edge linking and boundary detection techniques to connect broken edges and identify the boundaries of different regions. Explain how these techniques help distinguish agricultural fields and improve the accuracy of land-region identification.

Comprehensive Solution & Engineering Analysis:

1. Region-Based Segmentation (Seed Region Growing & Split-and-Merge):

When aerial land cover regions (e.g., lush crops vs. soil) share homogeneous spectral signatures but lack sharp intensity gradients at boundaries, edge detection produces incomplete contours. Region-based segmentation groups pixels based on spatial similarity.

A. Seeded Region Growing (SRG):

Start with seed pixels (x_0, y_0) in representative land regions (e.g., crop field center, lake center).

Growing Criterion: Examine 8-connected neighbors (x, y). Append pixel to region R_k if:

$| I(x, y) - \mu_{R_k} | <= T_{region}$

Where μ_{R_k} is the updated mean intensity of region R_k, and T_region is similarity tolerance.

B. Quadtree Split-and-Merge Algorithm:

Split Step: If a square image region R has high intensity variance $\sigma_R^2 > T_{var}$ (indicating mixed land types), split R into 4 equal sub-quadrants R_1, R_2, R_3, R_4.

Merge Step: If adjacent sub-regions R_i and R_j satisfy predicate P(R_i U R_j) = TRUE ($|\mu_{R_i} - \mu_{R_j}| <= T_{merge}$), merge them into a single contiguous field polygon.

2. Boundary Repair via Edge Linking & Hough Transform:

Aerial shadows, tree canopy gaps, and dirt roads create broken boundary lines between crop fields. Two complementary edge-linking methods are applied:

A. Local Processing Edge Linking:

Two edge pixels p(x_1, y_1) and q(x_2, y_2) are linked into a continuous border if:

Magnitude Similarity: | E(p) - E(q) | <= T_E

Direction Similarity: | θ(p) - θ(q) | <= T_θ (within ±45°)

Spatial Proximity: Euclidean distance d(p, q) <= d_max.

B. Hough Transform for Straight Line Boundary Detection:

Collinear fence lines and canal borders are detected by mapping edge points (x, y) into polar parameter space (ρ, θ):

$\rho = x \cdot \cos(\theta) + y \cdot \sin(\theta)$

Accumulator array A(ρ, θ) receives votes for each edge point. Local peaks in A(ρ, θ) define complete, unbroken straight-line field boundaries bridging physical gaps.

Land Cover Category	**Region-Based Segmentation Role**	**Edge Linking & Boundary Detection Role**
Agricultural Crop Fields	Clusters homogeneous green vegetation pixels into uniform crop parcels	Hough Transform links straight fence and road boundaries
Water Bodies (Lakes/Rivers)	Groups dark, uniform water pixels based on low NIR reflectance	Local edge linking connects broken riverbank gradients
Bare Soil & Farm Tracks	Separates dry soil regions based on texture & intensity variance	Delineates narrow dirt track borders separating crop zones
Overall GIS Accuracy	Prevents over-segmentation within uniform crop canopies	Ensures precise vector polygon generation for yield estimation

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____